

7. Write a program to implement Logistic Regression (LR) algorithm in python

Aim

To implement **Logistic Regression** for classification using Python.

Software Required

- Python 3.x
- NumPy
- Scikit-learn

Procedure

1. Load the Iris dataset.
2. Select two classes for binary classification.
3. Split the data.
4. Create the Logistic Regression model.
5. Train the model.
6. Predict the test samples.
7. Calculate accuracy.

Code

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

iris = load_iris()
X = iris.data
y = iris.target

# Binary classification
X = X[y != 2]
y = y[y != 2]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
model = LogisticRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Predicted Values:")
print(y_pred)
print("\nAccuracy:",
      accuracy_score(y_test, y_pred))
```

Output:

```
... Predicted Values:
[1 1 1 0 0 0 0 1 0 0 0 0 1 0 1 0 1 1 0 0]

Accuracy: 1.0
```

Result

Thus, **Logistic Regression** was successfully implemented for classification.